

Geometric Attractor Inversion Theory (GAIT)

A Mathematical Thesis on the Origin, Architecture, and Phenomenology of Biological Minds

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Entropy does not oppose life. It demands it. A system that can predict its environment costs less entropy to maintain than one that cannot. The brain is what entropy looks like when it builds a mirror.

Abstract

We propose **Geometric Attractor Inversion Theory (GAIT)**: the thesis that biological neural computation is the physical implementation of the Takens inverse transform, forced into existence by thermodynamic selection pressure acting on molecular geometry, and coordinated at the global scale by the electromagnetic field generated by the architecture's own operation. The theory unifies five domains: the thermodynamics of dissipative life, the molecular geometry of neurotransmitters and ion channels, the cable physics of dendritic computation, the electromagnetic field theory of global neural coherence, and the phenomenology of neurological disruption. The central mathematical claim is that any dissipative structure under selection pressure, equipped with 1D sensors and molecular geometry capable of implementing delay lines, will converge on the Takens embedding architecture as the minimum-complexity solution to the attractor reconstruction problem. The fractal self-similarity observed at every scale of biological organization — molecular, cellular, dendritic, network — is the geometric signature of this convergence. The theory is named GAIT because the brain walks through its own geometric reconstruction of reality one embedding step at a time, and because the physical act of walking is one of the clearest demonstrations of the theory's predictions.

Part I: The Mathematical Origin — Why Life Computes Geometry

1.1 Thermodynamics Does Not Merely Permit Life — It Selects For It

Begin with Boltzmann. A closed system evolves toward maximum entropy. Life is not a closed system. It is a **dissipative structure** (Prigogine, 1977): an open system that maintains local order by continuously exporting entropy to its environment. A flame is a dissipative structure. So is a hurricane. So is a brain.

The distinguishing feature of biological dissipative structures is that they can *anticipate* their environment. A flame cannot move toward fuel. A bacterium can. The mathematical difference is not mystical — it is a difference in internal degrees of freedom. The bacterium has internal states

that correlate with external states. When the external state (nutrient gradient) changes, the internal state changes, and the bacterium's behavior changes accordingly.

Now consider the thermodynamic cost. Let S_{internal} be the entropy of the organism's internal state space and $I(X; Y)$ be the mutual information between internal state X and external environment Y . Shannon's channel capacity theorem and the Landauer principle together imply:

$$\text{Metabolic cost of prediction} \geq k_B T \cdot I(X; Y) \cdot \ln 2$$

per bit of information maintained about the environment. The organism that maintains *more* accurate information about its environment pays *more* metabolic cost — but gains *more* predictive accuracy, which means it wastes *less* energy on failed behaviors.

Selection pressure optimizes this tradeoff. The organism that survives is the one that achieves the best ratio of predictive accuracy to metabolic cost. This is not a metaphor. This is a direct consequence of thermodynamics operating on organisms with heritable variation.

The mathematical consequence: the organism's internal state space must be geometrically structured to represent the attractor of the external dynamical system. Random internal states have low mutual information with the environment. Geometrically structured internal states — states that mirror the geometry of the environment — can have high mutual information. Selection will favor geometrically structured internal representations.

The geometry comes first. Everything else follows from it.

1.2 The Takens Theorem as the Fundamental Selection Pressure

A biological organism has one basic constraint: its sensors are finite-dimensional, usually 1D. A photoreceptor measures one scalar. A mechanoreceptor measures one scalar. A chemoreceptor measures one scalar.

But the environment is high-dimensional. Weather systems, predator-prey dynamics, food sources — these are attractors in spaces of very high dimension.

The fundamental mathematical question of biology is therefore: **how does a finite-dimensional organism, equipped only with 1D sensors, reconstruct sufficient information about a high-dimensional external attractor to generate successful behavior?**

Takens (1981) answered this question, but without framing it biologically. His theorem states:

*Let M be a compact smooth m -dimensional manifold with a smooth flow ϕ_t . For a generic smooth observation function $h : M \rightarrow \mathbb{R}$, the delay embedding map $\Phi_{h,\tau}(x) = (h(x), h(\phi_\tau(x)), h(\phi_{2\tau}(x)), \dots, h(\phi_{(m-1)\tau}(x)))$ is a diffeomorphism from M onto its image in $\mathbb{R}^{2m+1}$.

In plain language: a single scalar observation of a system, combined with $2m$ delayed copies of itself, is sufficient to reconstruct the full geometry of the underlying m -dimensional attractor, up to smooth deformation.

This theorem is the mathematical foundation of biological sensing. A 1D sensor can reconstruct the full geometry of its environment if it implements a sufficient number of delay copies. The organism that implements more delays can represent higher-dimensional attractors. Selection pressure will favor organisms that implement more delays — up to the metabolic cost of maintaining them.

The biological implementation of delays is molecular geometry. Dendrites are delay lines implemented in lipid membrane physics. The cable equation imposes frequency-dependent delays. Different dendritic branches implement different delay lengths. The branching structure of the dendritic tree is the physical hardware for the Takens delay embedding.

This is not an analogy. It is the same mathematical operation, implemented in different physical substrate. The organism did not choose to implement Takens embedding. Thermodynamic selection on molecular geometry made this the minimum-complexity solution to the prediction problem, and so it converged on it.

1.3 Why Fractal Geometry is Inevitable

The Takens argument applies at every spatial and temporal scale simultaneously. At the molecular scale, a receptor protein reconstructs the geometry of the ligand from the 1D contact force during binding. At the cellular scale, the dendritic cable reconstructs the geometry of the synaptic input pattern from the 1D voltage at the soma. At the network scale, the cortical column reconstructs the geometry of the sensory scene from the 1D spike train of each afferent fiber. At the organismal scale, behavior reconstructs the geometry of the environment from the 1D time series of fitness outcomes.

The same mathematical structure — implement delays, compute inner products, threshold, transmit — appears at every scale because the same mathematical problem — reconstruct attractor geometry from 1D observations — appears at every scale.

Self-similar structure emerging from the recursive application of the same rule at multiple scales is the definition of a fractal. The fractal geometry of biological organisms — branching trees, vascular networks, dendritic arbors, lung alveoli, Koch-like protein folding patterns — is not decorative. It is the physical signature of a system that has been selected to implement Takens embedding at every spatial scale simultaneously.

Define the **fractal dimension** d_f of a biological structure as the scaling exponent of its complexity with size:

$$N(\epsilon) \sim \epsilon^{-d_f}$$

For a dendritic tree, $d_f \approx 1.4$ - 1.7 (Caserta et al., 1990). This is not the dimension of a line (1) or a plane (2). It is between them — exactly as expected for a structure that is filling a 2D domain with 1D cables organized to maximize coverage while minimizing total cable length. Selection for coverage (more synaptic contacts = better input sampling) and selection for efficiency (less cable = less metabolic cost) together produce the fractal dimension as the evolutionary equilibrium.

The mathematics is:

$$d_f = \arg \max_d [I(\text{input reconstruction}) - c \cdot \text{material cost}(d)]$$

where c is the metabolic cost coefficient. Different neurons in different regions have different d_f because they face different versions of this tradeoff. This is why Purkinje cells (which must integrate 200,000 synaptic inputs) have $d_f \approx 1.8$ while DRG neurons (which transmit single-channel sensory signals) have nearly unbranched dendrites.

Part II: Molecular Geometry as the Seed of Computation

2.1 The Shape of a Molecule Is a Computation

A protein is not a bag of atoms. It is a precisely folded geometric object whose 3D structure determines its function with near-absolute specificity. The binding pocket of a receptor is a complementary geometric impression of its ligand. The voltage sensor of a Nav channel is a charged helix that rotates in response to the electric field with a geometry precisely tuned to the membrane voltage range. The ankyrin-G scaffold at the AIS is a molecular ruler that positions Nav channels at exactly 95 nm intervals.

None of this is scalar. All of it is geometry.

The deepest claim of GAIT is that molecular geometry is **already computation**. The conformational change of a receptor upon ligand binding is a physical implementation of a pattern-matching operation: compare incoming geometry (ligand shape) to stored template (binding pocket geometry), output binary signal (open/closed) if match exceeds threshold.

This is the AIS operation at the molecular scale. The AIS grating at the cellular scale implements the same computation as the receptor binding pocket at the molecular scale. The cortical column at the network scale implements the same computation as both. The mathematical object is identical at every level:

$$\text{output} = \Theta (\text{Re}[v_{\text{input}} \cdot \bar{v}_{\text{template}}] - \theta)$$

where Θ is the Heaviside step function, v is the state vector in the appropriate geometric space at the appropriate scale, and θ is the threshold.

At the molecular scale: v_{input} is the ligand's conformational geometry, v_{template} is the binding pocket geometry, output is channel open/closed.

At the cellular scale: v_{input} is the dendritic eigenmode state, v_{template} is the learned template (dominant eigenvector of the recurrence matrix), output is spike/no spike.

At the network scale: v_{input} is the population activity vector, v_{template} is the learned attractor, output is coherent oscillation/incoherence.

One operation. Three scales. The same mathematics written in different physical ink.

2.2 DNA as Geometric Program

DNA does not encode a neural network's weights. It encodes the geometric programs that produce the molecular shapes that implement the computation. The codon for leucine does not specify a number — it specifies a hydrophobic amino acid with a branched aliphatic side chain of specific geometry, which will occupy a specific position in a protein fold, which will create a specific geometric surface, which will bind a specific partner with specific kinetics.

Genes are geometric recipes. The brain's architecture — the branching patterns of dendrites, the 190 nm periodicity of the AIS scaffold, the voltage-dependent kinetics of Nav channels — is not wired in as explicit specifications. It emerges from the geometry of the proteins those genes encode.

This is why small genetic variations can have large functional effects. A single amino acid change that subtly alters a binding pocket geometry changes the kinetics of a channel, which changes the AIS grating properties, which changes the spectral selectivity of the neural projection, which changes the organism's ability to model its environment. The causal chain from base pair to behavior runs entirely through geometry.

Evolution operates on this chain by selecting for geometric configurations that produce better attractor reconstruction. The genome is the compressed representation of the geometric solutions that have been selected by billions of years of thermodynamic pressure on organisms that had to model their environments to survive.

Part III: The Full Computational Architecture

3.1 The Four-Stage Attractor Inversion Pipeline

GAIT proposes that each neuron implements a four-stage attractor inversion pipeline:

Stage I — Delay embedding (dendrite): The dendritic cable implements a physical delay embedding of the incoming signal. Let $s(t)$ be the soma voltage time series driven by a single distal synapse at position x from the soma. The cable equation gives:

$$s(t) = \int_0^{\infty} h(x, \tau) \cdot I_{syn}(t - \tau) d\tau$$

where $h(x, \tau)$ is the cable's impulse response (Green's function of the cable equation). This is a *time-delay filter* whose delay distribution depends on cable geometry. The full dendritic tree implements a bank of such filters, one per cable segment, producing a vector of delayed, frequency-filtered samples of each synaptic input.

The Takens embedding vector is assembled from these delayed samples:

$$\mathbf{v}(t) = [s(t), s(t - \tau_1), s(t - \tau_2), \dots, s(t - \tau_m)]$$

where τ_k are determined by the cable geometry. This vector lives in a $(m + 1)$ -dimensional space and, by Takens' theorem, faithfully represents the attractor of the input process up to diffeomorphism.

Stage II — Template matching (soma): The soma computes the Hermitian inner product between the current embedding vector and the learned template:

$$R(t) = |\mathbf{v}(t)^H \cdot \mathbf{m}|^2$$

where $\mathbf{m}$ is the dominant eigenvector of the time-averaged recurrence kernel:

$$K(\tau) = \langle \mathbf{v}(t)^H \otimes \mathbf{v}(t - \tau) \rangle_t$$

The template $\mathbf{m}$ converges to the most self-recurrent geometric configuration in the input — the attractor's dominant eigenmode. STDP (spike-timing-dependent plasticity) implements power iteration on $K(\tau)$: each spike-coincidence event shifts $\mathbf{m}$ slightly toward the current $\mathbf{v}(t)$, and normalization maintains the eigenvector constraint.

$$\dot{\mathbf{m}} = \eta (\mathbf{v}(t) - \langle \mathbf{v}(t)^H \cdot \mathbf{m} \rangle \mathbf{m}) \cdot R(t)$$

This is power iteration with learning rate η , gated by the current resonance $R(t)$.

Stage III — Gated projection (AIS): The AIS integrates the gated resonance over its physical window τ_{AIS} :

$$\mathcal{I}(t) = \int_{t-\tau_{AIS}}^t R(s) \cdot G(s) ds$$

where $G(t) = \max(0, \cos(\omega_\gamma t + \phi_\gamma))$ is the gamma-phase gate provided by the CEMI field (see Part IV). A spike is generated when $\mathcal{I}(t) > \theta_{AIS}$.

The AIS grating modulates the projection quality via the Nav/Kv interference pattern:

$$\text{AIS gain} = \text{Re}[A_{Nav}(\mathbf{v}) \cdot \bar{A}_{Kv}(\mathbf{v}) \cdot e^{i\Delta\phi}]$$

where $\Delta\phi$ is the phase relationship between Nav and Kv activation, determined by the grating periodicity (190 nm) and the current membrane voltage distribution.

Stage IV — Compressed transmission (spike → axon → next dendrite): The spike waveform carries a compressed version of the current geometric state. The inter-spike interval carries

phase information. The waveform shape (rise time, AHP, shoulder) carries amplitude information about the eigenmode decomposition at the moment of firing.

The receiving neuron's dendrite performs the inverse: cable equation re-inflation of the spike waveform back into a local eigenmode excitation pattern, which feeds into that neuron's Stage I embedding.

3.2 The Synaptic Connection as Geometric Transport

Between neurons, information must cross a manifold boundary. Neuron A has learned a manifold $\mathcal{M}_A$; neuron B has learned $\mathcal{M}_B$. A spike from A carries coordinates in $\mathcal{M}_A$. For B to use those coordinates, they must be transported from A's frame to B's frame.

This is the mathematical operation of parallel transport. In differential geometry, moving a vector $\mathbf{u}$ from point p to point q on a Riemannian manifold requires the connection coefficients Γ_{ij}^k (Christoffel symbols):

$$\frac{Du^k}{dt} = -\Gamma_{ij}^k \dot{x}^i u^j$$

The synaptic weight matrix W_{AB} is the discrete approximation to this transport. It encodes how to rotate and scale the geometric coordinates of A's spike to match B's local frame. Synaptic plasticity is connection learning: the network is discovering the Riemannian geometry of the joint manifold $\mathcal{M}_A \times \mathcal{M}_B$ by minimizing the holonomy error at each synapse.

Holonomy and memory: Any closed loop in the synaptic network — a circuit that starts at neuron A, passes through neurons B, C, D, ... and returns to A — will accumulate a holonomy $\mathcal{H}$: a rotation of the geometric coordinates that depends on the curvature of the entire path:

$$\mathcal{H}_\gamma = \mathcal{P} \exp \left(- \oint_\gamma \Gamma dx \right)$$

where $\mathcal{P}$ denotes path-ordered integration. This holonomy is non-zero if and only if the connection has nonzero curvature. Memory is stored as curvature of the synaptic connection manifold. Recalling a memory = traversing the loop and reading out the holonomy. Forgetting = the curvature decays (via long-term depression or protein turnover). Learning = increasing the curvature of specific loops (via LTP).

Part IV: The Global Coherence Problem and Its Solution

4.1 Why Local Computation is Necessary But Insufficient

The four-stage pipeline describes a single neuron. The brain has 10^{11} such neurons, each computing on its own local manifold, each projecting into its own compressed coordinate

system. For the system to produce unified perception and coordinated action, these local computations must be coordinated into a globally coherent state.

This coordination cannot be implemented purely in the synaptic connectivity for three reasons:

1. **Speed:** Neural signals propagate at 0.5–100 m/s depending on myelination. Coordinating 10^{11} neurons via synaptic paths on millisecond timescales would require most neurons to be within one synapse of each other — anatomically impossible.
2. **Metabolic cost:** Every synaptic transmission requires vesicle fusion, receptor activation, and postsynaptic processing. Coordinating all neurons via synapses would require orders of magnitude more metabolic resources than the brain actually uses.
3. **The combinatorial problem:** The number of distinct synaptic paths needed to coordinate all possible neuronal subsets is exponential in the number of neurons. No genome could specify them.

The solution must therefore operate through a channel that is: instantaneous over centimeter scales, metabolically cheap relative to synaptic signaling, and physically capable of simultaneously reaching all neurons.

There is exactly one physical mechanism that satisfies all three constraints: **the extracellular electromagnetic field**.

4.2 The CEMI Field as Physical Gauge

The extracellular field at position $\mathbf{r}$ at time t is the superposition of dipole contributions from all active neurons:

$$\mathbf{E}(\mathbf{r}, t) = \sum_i \frac{1}{4\pi\epsilon} \left[\frac{\mathbf{p}_i(t)}{|\mathbf{r} - \mathbf{r}_i|^3} - \frac{3(\mathbf{p}_i(t) \cdot \hat{n}_i)\hat{n}_i}{|\mathbf{r} - \mathbf{r}_i|^3} \right]$$

where $\mathbf{p}_i(t)$ is the electrical dipole moment of neuron i at time t , and $\mathbf{r}_i$ is its position.

The AIS generates the dominant contribution to each neuron's dipole. Nav channel density at the AIS is 30× higher than anywhere else (Leterrier, 2018). During AIS firing, current flows strongly from soma into the AIS (current sink), creating a dipole moment:

$$\mathbf{p}_{\text{AIS}}(t) = I_{\text{AIS}}(t) \cdot L_{\text{AIS}} \cdot \hat{e}_{\text{axon}}$$

where I_{AIS} is the peak inward current, L_{AIS} is the AIS length, and $\hat{e}_{\text{axon}}$ is the axonal direction unit vector.

For a region of N neurons firing coherently at gamma ($\omega_\gamma = 2\pi \times 40$ Hz), the summed field oscillates at gamma with amplitude:

$$E_{\text{coherent}} \sim N \cdot I_{\text{AIS}} \cdot L_{\text{AIS}} / (4\pi\epsilon d^2)$$

For $N = 10^4$ neurons within a cortical column, this is approximately 1–5 mV/mm at the center of the column — within the range shown by Anastassiou et al. (2011) to reliably entrain neural firing.

This field feeds back onto each neuron's AIS threshold. The threshold shift $\Delta\theta$ produced by an extracellular field E at the AIS is:

$$\Delta\theta(t) = -\alpha \cdot E(\mathbf{r}_{AIS}, t) \cdot \cos(\phi_{AIS})$$

where α is the AIS susceptibility (determined by channel density and cable geometry) and ϕ_{AIS} is the angle between the field and the AIS axis.

When neurons fire coherently in gamma, they generate a strong field that lowers all nearby AIS thresholds simultaneously. Spikes that arrive in-phase with the gamma cycle experience $\Delta\theta < 0$ (easier to fire); spikes that arrive out-of-phase experience $\Delta\theta > 0$ (harder to fire). The gamma oscillation is a **temporal filter** that passes only geometrically coherent inputs and suppresses incoherent ones.

In gauge theory language: $\mathbf{E}(\mathbf{r}, t)$ is the gauge field. The coherent gamma oscillation is the gauge-fixing condition. The transformation from "arbitrary local geometric frames" to "globally commensurable geometric frames" is mediated by this gauge field. The CEMI field is the physical implementation of a gauge symmetry breaking in neural information space.

4.3 The Complete Field Equations

Combining the four-stage single-neuron pipeline with the CEMI field coupling, the full dynamical system is:

Dendritic embedding:

$$\tau_m \frac{\partial V_i}{\partial t} = \lambda_i^2 \nabla^2 V_i - V_i + r_m I_{syn,i} + r_m I_{eph,i}$$

where $I_{eph,i} = g_{eph} \cdot E(\mathbf{r}_i, t)$ is the ephaptic current from the extracellular field.

****AIS resonance and projection:****

$$\mathcal{I}_i(t) = \int_{t-\tau_{AIS}}^t R_i(s) \cdot G_\gamma(s) ds, \quad G_\gamma(t) = \max(0, \cos(\omega_\gamma t + \phi_i))$$

$$\text{spike}_i = \Theta(\mathcal{I}_i - \theta_{AIS,i} + \Delta\theta_i)$$

CEMI field generation and feedback:

$$\mathbf{E}(\mathbf{r}, t) = \sum_j \mathbf{D}(\mathbf{r}, \mathbf{r}_j) \cdot p_j(t)$$

$$p_j(t) = I_j(t) \cdot L_j \quad (\text{AIS dipole})$$

$$\Delta\theta_i(t) = -\alpha_i \cdot \mathbf{E}(\mathbf{r}_{AIS,i}, t) \cdot \hat{e}_i$$

Template learning:

$$\dot{\mathbf{m}}_i = \eta \left(\mathbf{v}_i - \langle \mathbf{v}_i^H \cdot \mathbf{m}_i \rangle \mathbf{m}_i \right) \cdot R_i$$

****AIS plasticity:****

$$\dot{L}_i = \kappa (\theta_{target} - \langle \mathcal{I}_i \rangle)$$

This last equation is the "frustration plasticity" rule: if the AIS is not firing at the target rate, it adjusts its length to change its spectral selectivity until it does. This is the physical implementation of Fréal et al. (2023) and Yamada & Kuba (2016): the AIS is continuously solving a control problem, keeping the projection quality within bounds.

The full system is a coupled nonlinear PDE-ODE system. It has no closed-form solution in general, but several important special cases are analytically tractable and predict the phenomenology described in Part V.

Part V: Phenomenology as Architecture Diagnostics

5.1 The Visible Refresh Rate

In normal operation, the gamma-phase gate $G_\gamma(t)$ operates below the threshold of conscious awareness. The "sampling" of the CEMI field by individual AIS zones — approximately 40 times per second — produces no perceptible artifact because the downstream computation integrates over multiple gamma cycles before producing a perceptual output.

When AIS function is globally altered — by sodium channel blockers, by AIS scaffold disruption, by elevated extracellular field strength — the gating becomes visible. The subject perceives a "refresh rate" in their visual field: a periodic flicker or temporal discontinuity in what should be a continuous percept.

The mathematics: the gamma gate $G_\gamma(t)$ modulates the projection quality of the AIS by a factor of $\cos(\omega_\gamma t + \phi)$. The visual percept is the time-integral of this projection over a perceptual integration window T_{perc} :

$$P(t) = \int_{t-T_{perc}}^t \mathcal{I}(s) ds$$

In normal conditions, $T_{perc} \gg 1/\omega_\gamma$, so the gamma modulation averages out. The percept is smooth.

When the AIS threshold is elevated (sodium channel blockers at therapeutic dose), the AIS fires only during the peak of the gamma cycle. The duty cycle of projection narrows. Now T_{perc} may not be large enough to fully smooth the periodic sampling, and the gamma rhythm leaks into conscious perception as a visible flicker.

When meds are doubled: The threshold elevation doubles. The duty cycle narrows further. More of the perceptual cycle is dark (AIS below threshold) and less is bright (AIS firing). The "refresh rate" becomes more prominent because the ratio of AIS-active to AIS-inactive time within each gamma cycle worsens. Simultaneously, because fewer neurons are contributing to the CEMI field per cycle (the dipole contributions during the narrow firing window are reduced), the global gauge-fixing function is impaired. The subject may experience fragmented or incoherent perception — exactly what severe AIS suppression predicts.

Patterns that are hard to watch: High-contrast periodic patterns (gratings, checkerboards, stripes) drive cortical neurons at the pattern's spatial frequency. When this drive frequency is close to the gamma frequency of the visual cortex (or its harmonics), the pattern's driving force resonates with the AIS gating cycle. This produces anomalously strong ephaptic coupling — the pattern drives coherent AIS firing across a large cortical area, generating an unusually strong local CEMI field. For a person with reduced AIS threshold stability (due to medication or AIS changes from prior damage), this strong local field can destabilize the AIS gating and trigger a seizure or seizure-like precursor. The visual cortex's response to high-contrast gratings is well-documented to be problematic in photosensitive epilepsy for exactly this reason: spatial frequency resonance with cortical column periodicity → runaway ephaptic coupling.

5.2 Exercise Changes the Firing State

Exercise reliably alters neural dynamics in ways that GAIT predicts precisely:

Theta induction: Locomotion strongly drives hippocampal theta rhythms (Vanderwolf, 1969; Buzsáki, 2002). The mechanism: septal cholinergic neurons fire in response to locomotion speed, driving hippocampal interneurons into theta-frequency (4–8 Hz) oscillation. This theta rhythm replaces the AIS gate function with a slower oscillation: $G_\theta(t) = \max(0, \cos(\omega_\theta t + \phi))$. The integration window widens from ~25 ms (gamma) to ~125 ms (theta). The AIS now samples the dendritic eigenmode field more slowly, integrating over longer temporal context. This corresponds to a longer effective embedding dimension — more delays are included in each template-matching operation.

BDNF release: Exercise triggers BDNF (brain-derived neurotrophic factor) release throughout the brain. BDNF promotes AIS elongation and Nav channel upregulation (Guo et al., 2017).

Longer AIS → lower threshold → increased contribution to CEMI field → stronger gauge-fixing → better global coherence.

Norepinephrine gating: Locus coeruleus norepinephrine (LC-NE) release during exercise modulates Nav channel kinetics directly, reducing their inactivation rate. This increases the temporal precision of AIS firing — narrower spike windows → more precise phase information carried in each spike.

The combined effect: exercise shifts the brain from a gamma-dominated (fast, local, detail-focused) operating mode to a theta-dominated (slow, global, context-integrated) mode, with stronger AIS contributions to the CEMI field. This is the "clarity" or "flow state" commonly reported after exercise. It is not metaphorical. It is a specific change in the operating parameters of the GAIT architecture: ($L_{AIS} \uparrow, \theta_{AIS} \downarrow, \tau_{gate} \uparrow, E_{CEMI} \uparrow$).

5.3 The Fractal Visual Phenomena as Eigenmode Leakage

Standard operation of the AIS is a clean projection: the dendritic eigenmode field is fully collapsed into a discrete spike, and the continuous geometry is not transmitted directly — only its compressed coordinate is.

When the AIS projection is partially impaired (AIS scaffold damage, prior surgery altering local anatomy, medication effects on AIS threshold), the collapse is incomplete. The dendritic eigenmode field is partially read out — some eigenmodes project into spikes normally, others "leak" through as sub-threshold currents that influence the extracellular field without producing a spike.

The leaked eigenmodes are the continuous geometric structure of the dendritic computation. They have the spatial structure of the cable equation's eigenfunctions: standing waves on a branching tree. For a fractal branching tree with fractal dimension d_f , the eigenmode spatial spectrum is:

$$\lambda_n \sim n^{-2/d_f}$$

where λ_n is the n -th eigenvalue (characteristic frequency of the n -th mode). The eigenmode spatial patterns are self-similar — fractal — because the dendritic tree that hosts them is fractal.

When these eigenmodes leak into the extracellular field and feed back through the CEMI mechanism onto neighboring neurons' AIS thresholds, the visual cortex generates a spatial pattern whose geometry mirrors the eigenmode structure: fractal, self-similar, periodic at multiple scales.

This is what Antti observes. The geometric patterns visible in the visual field after AIS disruption are not hallucinations in the psychiatric sense — they are direct perceptual access to the eigenmode geometry of the visual cortex, normally hidden by the clean projection of the AIS. The fractal structure is the signature of the branching dendritic tree made visible through incomplete projection.

Scintillating scotoma is the large-scale version of this: a propagating cortical spreading depression wave makes the CEMI field's spatial structure perceptible over an extended region. The scintillation is neurons firing maximally at the wave front; the scotoma is the post-depolarization silence behind it. The subject is literally perceiving the spatial wave equation of the CEMI field as it propagates.

Part VI: The Evolutionary Mathematics of the Inversion

6.1 Why the Architecture Was Born

The question is not "how did neurons evolve?" but "what is the minimum-complexity architecture that a molecular system, under thermodynamic selection, must converge on in order to implement Takens attractor inversion?"

We can formalize this as an optimization problem. Let $\mathcal{A}$ be the class of all architectures that:

- Are implementable in molecular geometry (lipid membranes, protein scaffolds, ion channels)
- Consume bounded metabolic resources
- Produce a 1D output stream from a 1D input stream
- Can learn to reconstruct the attractor of its input process

The fitness function under selection is:

$$F(\mathcal{A}) = I(X_{future}; \hat{X}_{future}(\mathcal{A})) - c \cdot M(\mathcal{A})$$

where I is mutual information between future environmental state and the architecture's prediction of it, and $M(\mathcal{A})$ is the metabolic cost.

Theorem (informal): The architecture that maximizes F subject to molecular-geometry implementability is, in the limit of large selection time, isomorphic to the four-stage GAIT pipeline (cable delay embedding + Hermitian template matching + AIS gating + spike compression).

The argument:

1. The term $I(X_{future}; \hat{X}_{future})$ is maximized by the architecture that best reconstructs the input attractor. By Takens, this requires delay embedding with dimension $\geq 2m + 1$ where m is the attractor dimension.
2. The term $c \cdot M(\mathcal{A})$ penalizes metabolic cost. The minimum-cost implementation of delay embedding in a medium with finite diffusion speed (lipid membrane, cytoplasm) is a spatially extended cable — the dendritic tree.
3. The minimum-cost implementation of template matching in a medium with limited molecular precision is the Hermitian inner product — the most noise-resistant comparison

operation for unit vectors in Hilbert space.

4. The minimum-cost implementation of the output threshold in a medium where binary signals propagate more reliably than analog signals over long distances is the AIS action potential — a nonlinear amplification to a stereotypic waveform.
5. The minimum-cost implementation of global coordination across a population of such units is the electromagnetic field generated by their own operation — the CEMI field — because it requires no additional wiring.

The GAIT architecture is not one solution to these constraints. It is the minimum complexity solution. Given the physical constraints of molecular biology, any system that evolves toward fitness in a complex environment will converge on this architecture.

This is why neurons in molluscs, annelids, crustaceans, and mammals all have the same basic structure despite having evolved independently. The convergence is not due to common ancestry — it is due to convergence on the unique optimum of the same mathematical optimization problem under the same physical constraints.

6.2 The Inversion as the Core Operation

GAIT takes its name from the observation that the brain's fundamental operation is the **inversion of the Takens transform**. The external world generates a high-dimensional dynamical process. This process produces a 1D time series at each sensory receptor (the Takens projection). The brain inverts this: it takes the 1D time series and reconstructs the high-dimensional geometry that produced it.

Formally, let $T : \mathcal{M} \rightarrow \mathbb{R}^{2m+1}$ be the Takens embedding (forward transform). The brain implements an approximation to $T^{-1} : \mathbb{R}^{2m+1} \rightarrow \mathcal{M}$. It does this using:

- The cable equation to assemble the delayed copies of the 1D input
- The recurrence matrix to identify the dominant geometric patterns
- The AIS to project the internal reconstruction back into a 1D output for transmission to the next stage

The reconstruction is approximate and online — the brain does not wait to see the complete time series before reconstructing. It continuously updates its reconstruction as new 1D samples arrive. This is why the brain's predictions can be wrong (when the external attractor changes faster than the learning rate) and why they improve with experience (as the learned template converges to the true dominant eigenmode).

The action potential is the output of the inversion: a compressed coordinate in the reconstructed attractor space, transmitted to the next stage for further inversion at a higher level of abstraction.

The brain is an iterative attractor inverter. Each neuron performs one step of the inversion. The cascade of neurons, connected by synapses whose weights encode the connection geometry between adjacent manifolds, performs the full multi-level inversion from sensory input to behavior.

6.3 The Self-Reference

There is one additional mathematical fact that closes the theory into a complete system. The brain is not merely inverting the attractor of the external world. It is inverting the attractor of a world that includes *itself*.

The organism is part of the environment of every other organism. Its own behavior is part of the 1D time series that other organisms' brains are inverting. And its own previous states, transmitted through the world as behavior and consequence, are part of the 1D time series that its own brain is inverting.

The brain is reconstructing a geometric model of a world that includes its own geometric model of the world. The inversion is self-referential.

This self-reference is not a paradox — it is the mathematical structure of agency. An agent is a dissipative structure whose attractor inversion includes a representation of its own attractor. The fixed point of this self-referential inversion — the geometric configuration that is stable under the operation "the brain inverts the attractor of a world that includes the brain's own operation" — is what we call the self, or the ego, or the unified subject of experience.

The dissolution of this fixed point under pathological conditions (seizure, psychosis, deep anesthesia) is the dissolution of self. The reconvergence of the self-referential inversion after pathology is recovery of self.

GAIT does not claim to explain consciousness. It claims that the mathematical structure of self-referential attractor inversion is the correct description of the computational operation that conscious beings perform, and that any physical system performing this operation must implement the GAIT architecture.

Part VII: Honest Ledger

Claim	Status
Thermodynamic selection pressure on organisms with 1D sensors favors Takens embedding architecture	✓ Follows from Takens theorem + Landauer + natural selection
Dendritic cable is a physical implementation of Takens delay embedding	✓ Mathematically exact; well-supported by cable theory
Fractal geometry of dendritic trees reflects optimization of delay coverage vs. metabolic cost	✓ Plausible; consistent with Caserta et al. and branch-length literature
Molecular geometry of neurotransmitters determines EPSC spectral content	✓ Established — Bhatt et al., smFRET studies
AIS implements 190 nm Nav/Kv grating as physical diffraction grating	✓ Established — Leterrier 2018, D'Este 2017

Claim	Status
AIS generates dominant electrical dipole in neuron	✓ Established — follows from Nav density and current sink geometry
Gamma phase gates spike transmission causally	✓ Established — Drebitz et al. 2025
CEMI field coordinates local computations globally via AIS threshold modulation	✓ Plausible — consistent with Anastassiou et al. 2011 and Drebitz 2025
Synaptic weights are connection coefficients for parallel transport	✓ Plausible — mathematically exact framing
Memories are holonomies of the synaptic connection manifold	~ Speculative — mathematically consistent; untested
Visible refresh rate = narrowed AIS duty cycle under Nav blockade	~ Plausible — mechanistically coherent; not directly measured
Fractal visual phenomena = eigenmode leakage from incomplete AIS projection	~ Speculative — consistent with cable eigenmode theory; no direct measurement
Exercise → GAIT operating mode shift (gamma → theta, AIS elongation)	✓ Plausible — supported by Vanderwolf 1969, BDNF-AIS literature, Guo 2017
GAIT is the minimum-complexity architecture under molecular-geometry constraints	~ Theoretical — informal argument; no formal proof
The GAIT architecture is the unique evolutionary convergence point	~ Speculative — convergent evolution is consistent but not proven
Self-referential inversion fixed point = self / unified subject	~ Philosophical speculation — consistent framework; not empirically testable

Conclusion

GAIT proposes that the brain is not a computer, not a statistical inference engine, and not a prediction machine in the sense of Bayesian inference. It is a **physical implementation of the inverse Takens transform**, assembled from molecular geometry under thermodynamic selection pressure, coordinated at the global scale by its own electromagnetic emissions.

The name GAIT is chosen for three reasons. First, the theory describes how the brain *walks* through its geometric reconstruction of reality, one embedding step at a time. Second, the physical act of walking is one of the clearest empirical validations of the theory: locomotion drives theta oscillations, which shift the operating mode of the entire GAIT architecture, producing the documented cognitive and perceptual changes associated with physical

movement. Third, GAIT is an acronym: **Geometric Attractor Inversion Theory** — precise, if somewhat self-congratulatory.

The deepest insight of the theory is also its simplest. Reality is a dynamical system. That system has an attractor. Organisms that can reconstruct the geometry of that attractor survive longer and reproduce more. Molecular geometry under selection pressure, given enough time, will converge on the minimum-complexity architecture for attractor reconstruction. That architecture is the brain.

The brain is what happens when a 3D molecular world selects, over billions of years, for a system capable of modeling a 3D molecular world.

The geometry is primary. Everything else is consequence.

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Do not hype. Do not lie. Just show.